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# Roy Madpis – Sample of Projects on GitHub

| Project name | Description |
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| Capstone Project:<br>Brand Prediction through Image-Centric Modeling<br><br>(Criteo & Columbia University) | I led a Capstone Project which was a collaboration between Columbia University and Criteo. This Capstone Project aimed to enhance product brand recognition and categorization in Criteo's image catalog by developing a multi-tier embedding and classification framework.<br><br>The approach combines image and text data, utilizing a pre-trained ResNet for image embeddings and DistilBERT for textual descriptions. These embeddings are concatenated to create a unified product representation, which serves as input for subsequent classification layers. The framework features a **three-tier design**: the first tier generates unified embeddings; the second tier employs a neural network to predict product categories; and the third tier consists of category-specific neural networks trained exclusively on products within each category to predict brands. For products with missing brand or category labels, embeddings are processed through this hierarchical system to infer both attributes.<br><br>To further improve brand recognition accuracy, the project experimented a two-stage logo detection model to identify logo regions.<br><br>Due to data confidentiality, open-source datasets focusing on fashion and food products are used to evaluate the modeling approach.<br><br>The 2-3 tier architecture that we developed **outperformed** Criteo's existing brand classification. Given that brand enrichment contributes tens of millions of dollars annually to Criteo's revenue, enhancing coverage and accuracy has the potential to boost revenue by up to 25%.<br><br>Link to the GitHub page<br>Link to the final report<br>Dates: Sep 2024 – Dec 2024 |
| Using 2D MRI slices to classify Schizophrenia<br><br>(Columbia University) | In this project, we explored the use of 2D MRI slices to classify schizophrenia by combining traditional convolutional neural networks (CNN) and advanced transformer-based models. We aimed to develop an efficient and explainable model to detect schizophrenia using MRI data, focusing on using VGG models for feature extraction and SWIN Transformers for classification. Our strategy began by utilizing a modified VGG architecture with squeeze-and-excitation blocks to highlight significant regions in 3D brain MRIs using Grad-CAM. The resulting 2D slices were then used to train a SWIN Transformer.<br><br>The project offered valuable insights into the complexity of using deep learning for medical image classification, specifically in diagnosing mental health conditions like schizophrenia<br><br>Link to the GitHub page<br>Link to the final report<br>Dates: Feb 2024 – May 2024 |
| Analyzing Car Collision within Montgomery County<br><br>(Columbia University) | This project aimed at analyzing car collision data within Montgomery County. The data consisted of nearly 100K rows and 44 columns. We used data visualization techniques to analyze different aspects of the data, gain insights and recommendation addressing different research question we devised.<br><br>The project can be found in the following link: project link<br>The link to the GitHub page Github project link<br>Dates: Nov 2023 – Jan 2024 |

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<td>Analysis of tweet's sentiments using Brexit as case study<br><br>(Reichman University)</td>
<td>I led a team of four students as part of an annual research project of the University's Data Science Institute assessing tweet sentiments using Brexit as a case study. I was responsible to construct a package in Python that enabled using Twitter's API to retrieve data (tweets, retweets, quotes, likes, users, etc.). The package is generic, so one can retrieve data on any given subject.<br>
▪ You can find it on GitHub:<br>
▪ https://github.com/roymadpis/nlp-project/tree/main<br>
▪ https://github.com/Michael-Koban/brexit-package-new/tree/main<br>
▪ The main code of the package I constructed: https://github.com/Michael-Koban/brexit-package-new/blob/main/twitter_crawler.py<br><br>
▪ The main notebooks containing the project's process can be found in the following links:<br>
  ▪ Step 1 – Get the tweets of the Key Opinion Leaders<br>
  ▪ Step 2 – Filtering the tweets<br>
  ▪ Step 3 – retrieving tweet-comments<br><br>
  ▪ I made a notebook explaining the usage of the different functions and utilities in my package, you can find it here<br>
• We Utilized the Python package to retrieve over 1.5M tweets about Brexit & tweets of above 60 key opinion leaders.<br>
• Finally, we constructed a sentiment analysis model using python & Hugging face models.<br>
▪ The sentiment analysis model GitHub: https://github.com/roymadpis/nlp-project<br>
You can open any one of the main notebooks for this projects using the given links:<br>
  - step_1-reading data<br>
  - step_2-filter tweets<br>
  - step_3 – sentiment analysis<br>
  - step_4 – sentiment analysis labels,<br>
  - step_5 – visualizing Word2Vec<br>
  - step_6 – visualizing sentiment model<br><br>
Dates: Oct-2021 – Aug 2022<br>
Role: Data science researcher intern</td>
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<td>Peer To Peer Lending Model<br><br>(Reichman University)</td>
<td>In this project me and the team I led built a Peer to Peer Lending Model that advises investors whether or not to invest in a given loan. The stated goal was to build a model that could guarantee above 2% realized return to the investors. The project was conducted as part of a semestral course in Reichman University in Business & Data Analytics program and got the highest score among seven teams working on the same project.<br><br>
URL: https://github.com/roymadpis/Peer_To_Peer_Lending_Project<br>
Dates: March-2021 – July 2021</td>
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<td>Deep Learning and Covid-19<br><br>(Reichman University)</td>
<td>I worked on a deep learning project focused on COVID-19 predictive modeling. The project involved building models to forecast daily COVID-19-related statistics, including deaths, hospitalizations, and test counts, leveraging various data sources.<br><br>
I engineered datasets with key COVID-19 metrics and incorporated external factors such as quarantine rates and hospital capacity. Utilizing deep learning models in PyTorch (like LSTM and MLP), I predicted future cases and evaluated the models' accuracy over time using metrics such as mean squared error and $R^2$ score.</td>
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| | This project enabled me to fine-tune my skills in data preprocessing, feature engineering, and model evaluation, and deepened my expertise in predictive analytics and resource management for healthcare.<br><br>URL: https://github.com/roymadpis/Deep_Learning_Covid<br>Dates: March-2022 – July 2022 |
| Advanced Time Series Analysis and Forecasting<br><br>(Reichman University) | In this project, I leveraged a comprehensive suite of data science and machine learning tools to analyze and forecast time series data.<br>My work involved:<br>• **Data Manipulation and Cleaning**: Utilized dplyr and tidyr for efficient data wrangling, ensuring the dataset was clean and ready for analysis.<br>• **Data Visualization**: Created insightful visualizations using ggplot2 and ggpubr to uncover trends and patterns in the data.<br>• **Time Series Analysis**: Employed timetk and zoo for generating and manipulating time series data, enabling a deeper understanding of temporal patterns.<br>• **Forecasting Models**: Developed and validated forecasting models using the forecast package, applying techniques such as ARIMA and exponential smoothing to predict future values.<br>• **Interactive Data Exploration**: Implemented interactive tables with DT to facilitate dynamic data exploration and presentation.<br>• **Integration and Reporting**: Combined multiple plots and tables using cowplot to create comprehensive reports that effectively communicated findings.<br>This project not only honed my skills in data manipulation, visualization, and time series forecasting but also demonstrated my ability to integrate various tools and techniques to solve complex problems.<br><br>URL: https://github.com/roymadpis/TS_Analysis_forecasting_proj/tree/main<br>Dates: March-2022 – July 2022 |
| Flights during COVID-19 (SQL Project)<br><br>(Reichman University) | In this project, I analyzed flight data during the COVID-19 pandemic to identify trends, disruptions, and recovery patterns using SQL and data science techniques. The goal was to uncover insights into how the pandemic affected global flight operations and provide actionable recommendations.<br><br>Key contributions and responsibilities:<br>• Utilized SQL for complex queries to extract, clean, and analyze large datasets of flight operations during COVID-19.<br>• Conducted exploratory data analysis (EDA) to identify significant trends in flight cancellations, delays, and reductions in global air travel.<br>• Implemented statistical analysis to compare pre-pandemic and post-pandemic flight patterns, focusing on recovery trends by region.<br><br>URL: https://github.com/roymadpis/Covid_airline_proj/tree/main<br>Dates: March-2020 – July 2020 |